Albert Zhou

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EDUCATION

Carnegie Mellon University | Mountain View, CA Aug 2026 – Jan 2028
Master of Software Engineering (MSE) (Expected)

University of Warwick | Coventry, UK Sep 2023 – Jun 2026
BSc in Computer Science, First Class Honours

- ***Relevant modules**: Computer Architecture, Artificial Intelligence, Machine Learning, Computer Graphics, Algorithms, Compiler Design, Operating Systems and Computer Networks, Database Systems*

INTERNSHIP EXPERIENCE

Amazon | SDE Intern | London, UK Jun 2025 – Sep 2025

- Engineered an agentic AI assistant for the AWS Billing team to autonomously troubleshoot service tickets for an internal AWS service; highly commended by the team and adopted into their triage workflow.
- Integrated the agent directly with the existing ticketing system so it could act on tickets without engineers switching tools.
- Designed the agent to perform automated root cause analysis by querying metrics, logs, and observability tools via internal APIs.
- Deployed the agent on AWS using Lambda, CloudWatch, S3, EC2, and IAM.

National Grid ESO | Intern | Wokingham, UK Mar 2024 – Apr 2024

- Developed a program to process energy data (State of Charge, Metered MW, MWh) and built a model to optimise efficiency.

PROJECTS

AI Video Game Coaching Analysis Project (OWAI) | *github.com/64NDA/OWAI* Sep 2025 – Jun 2026

- Developed OWAI, a system that analyses video game footage using YOLO-based object detection and computer vision models.
- Extracted gameplay data and constructed a timeline of in-game events for performance analysis.
- Integrated the data into a RAG-based LLM pipeline for coaching advice, preferred over a baseline LLM in 87% of player ratings.
- Final year dissertation, awarded a First-Class mark; rated 4.8/5 in user testing.

NBA Match Predictor | *github.com/64NDA/NBA_Match_Predictor* Jul 2024 – Sep 2026

- Built a model predicting NBA games, combining machine learning with classic sports analytics (player impact ratings, Elo).
- Reached 67.1% accuracy over four seasons, beating an Elo-based baseline (64.8%).
- Created a Streamlit web app showing daily predictions that account for injury reports.

Traffic Flow Predictor | Software Engineering Coursework Feb 2025 – Apr 2025

- Led a team building a traffic flow predictor for a four-way junction, developing the simulation and desktop application.

RESEARCH EXPERIENCE

Ruijin Hospital R&D | Intern | Shanghai, China Jul 2024 – Oct 2024

- Prepared data for foundation model training and developed a program to format and process whole-slide image data.
- Evaluated foundation models for diagnostic assistance and education, analysing report generation and text-to-image recall.
- Co-authored a review of foundation models in pathology, assessing clinical readiness; accepted in Med (Cell Press).

SAFARI Research Group, ETH Zurich | Junior Intern | Zurich, Switzerland Aug 2023 – Sep 2023
Supervisor: Professor Onur Mutlu

- Conducted research on DRAM retention failure, designing and running experiments to evaluate how parameters such as refresh time intervals and temperature affect failure rates in memory.
- Used Linux systems to run experiments on DRAM Retention through DRAM-Bender, an FPGA-based memory controller.

PUBLICATIONS

Computational Pathology in the Era of Emerging Foundation and Agentic AI | Co-author Med (Cell Press), accepted
arxiv.org/abs/2603.05884

- Worked on the initial investigation, comparing existing foundation model solutions and evaluating their use cases in pathology.

Answering the Call to ARMs with PACER: Power-Efficiency in Storage Servers | Co-author MSST 2024
https://www.msstconference.org/MSST-history/2024/Papers/msst24-4.2.pdf

- Analysed power consumption and IO metrics in ARM-based, flash storage servers, focusing on optimising power efficiency.

SKILLS

Programming Languages: Python, C++, C, Java, Haskell, SQL, TypeScript, JavaScript, HTML/CSS
AI & ML: Agentic AI, LLM integration, Retrieval-Augmented Generation (RAG), PyTorch, TensorFlow, Hugging Face, OpenAI API, vector databases, computer vision (YOLO/Ultralytics, OCR)
Cloud & DevOps: AWS (Lambda, EC2, S3, CloudWatch, IAM, Bedrock), Microsoft Azure, CI/CD pipelines, Git
Databases & Web: PostgreSQL, SQLite, Node.js, Express, Streamlit
Systems & Tools: Linux, Arduino, Jetson Nano, Unreal Engine, Blender

ADDITIONAL

Languages: English (native), Mandarin (fluent), French (intermediate), Latin (beginner)
Interests: Basketball, Hockey, Badminton, Competitive Overwatch (university-level, top 8 finishes), Fine art, Digital art, Violin, Piano